

CYNThERA

Drug Repurposing AI Report

Drug: Aspirin

Disease: Pancreatic Cancer

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Recommendation: UNCERTAIN

Executive Summary

CYNThERA v2.0 analysis of Aspirin → Pancreatic Cancer produced a recommendation of 'UNCERTAIN'. Evidence Strength: 98.8% (HIGH) | Mechanistic Plausibility: 44.8% (LOW) | Risk Level: 54.2% (MEDIUM). 7 claim(s) extracted from literature (7 from openalex), 0 contradiction(s) detected. Prior knowledge: regulatory NONE (0%), repurposing EMERGING, mechanistic MODERATE, clinical HUMAN_EVIDENCE, knowledge maturity GROWING. Safety grade: C. 2 mechanistic path(s) traced.

Three-Dimensional Score Assessment

Dimension	Score	Level	Details
Support Score (SS)	0.988	HIGH	59 evidence records
Mechanistic Score (MS)	0.448	LOW	8 pathways
Risk Score (RS)	0.542	MEDIUM	1 failed trials, 0 contradictions

Therapeutic Direction Alignment (Phase 4D)

Overall Directional Alignment: SUPPORTS

Therapeutic alignment *SUPPORTS* hypothesis: Drug action on primary target(s) (PTGS2) is directionally concordant with disease-target therapeutic requirements.

Target	Drug Action	Desired Action	Alignment	Evidence Groups
PTGS2	INHIBITION	INHIBITION	SUPPORTS	1 supp / 0 opp groups

Scientific Context — Prior Knowledge

Dimension	Status	Confidence	Evidence
Regulatory	NONE	0%	No ChEMBL indication match found for this disease

Repurposing	EMERGING	50%	Pair-scoped literature exists for this disease
Mechanistic	MODERATE	45%	Validated path: Aspirin → PTGS2 (P35354) [inhibitor (ChEMBL)] → Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)
Clinical	HUMAN_EVIDENCE	50%	4 registered trial(s) for this pair: NCT04245644 (N/A, UNKNOWN), NCT06060606 (N/A, UNKNOWN)
Knowledge Maturity	GROWING	81%	Related cache entry: aspirin → colorectal cancer (similarity 0.81)

Related prior-knowledge pairs: aspirin → colorectal cancer (0.81), itraconazole → non-small cell lung cancer (0.21), metformin → cancer (0.15)

Candidate Biological Mechanisms

Candidate Mechanism 1: Mechanism 1: PTGS2 (P35354) via Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [**WEAK_SPECULATIVE**] [DIR: UNKNOWN]

Drug: Aspirin → Target: PTGS2 (P35354) → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) → Gene: TP53 → Disease: Pancreatic Cancer

- Drug: Aspirin —**INHIBITOR**→ Target: PTGS2 (P35354) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

- Target: PTGS2 (P35354) —**PARTICIPATES_IN**→ Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) —**CONTAINS_ASSOCIATED_GENE**→ Gene: TP53 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: TP53 —**ASSOCIATED_WITH**→ Disease: Pancreatic Cancer [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: PTGS2 (P35354) → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) → Gene: TP53 is not supported by a mapped claim.

Candidate Mechanism 2: Mechanism 2: PTGS2 (P35354) via Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [**WEAK_SPECULATIVE**] [DIR: UNKNOWN]

Drug: Aspirin → Target: PTGS2 (P35354) → Reaction: Expression of PTGS2 (R-HSA-6790029) → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) → Gene: TP53 → Disease: Pancreatic Cancer

- Drug: Aspirin —**INHIBITOR**→ Target: PTGS2 (P35354) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

• Target: PTGS2 (P35354) —**PARTICIPATES_IN_REACTION**→ Reaction: Expression of PTGS2 (R-HSA-6790029) [\[Open Reactome Reaction\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

• Reaction: Expression of PTGS2 (R-HSA-6790029) —**PART_OF**→ Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

• Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)
—**CONTAINS_ASSOCIATED_GENE**→ Gene: TP53 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

• Gene: TP53 —**ASSOCIATED_WITH**→ Disease: Pancreatic Cancer [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

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Missing evidence: Direct biological bridge for Reaction: Expression of PTGS2 (R-HSA-6790029) → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) → Gene: TP53 is not supported by a mapped claim.

Reaction-Enriched Mechanistic Evidence (Phase 3)

Reactome reaction evidence describes the target's molecular role within the reaction. It does not by itself establish therapeutic activation or inhibition. Direction defaults to UNKNOWN.

Target	Reaction / Event	Role	Pathway	Direction	Source
PTGS2 (P35354)	Expression of PTGS2 (R-HSA-6790029)	PARTICIPATES_IN	Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)	DIRECTION_UNCERTAIN	Reactome

Biomedical Data Sources Accessed

Source Name	Status	Direct Source Portal
ChEMBL	SUCCESS	[Open ChEMBL]
UniProt	SUCCESS	[Open UniProt]
Reactome	SUCCESS	[Open Reactome]
Open Targets	SUCCESS	[Open Open Targets]

DisGeNET	SUCCESS	[Open DisGeNET]
PubMed	SUCCESS	[Open PubMed]
Europe PMC	SUCCESS	[Open Europe PMC]
ClinicalTrials.gov	SUCCESS	[Open ClinicalTrials.gov]

Recommendation Rule Engine Output

- Rule 5 (UNCERTAIN): Mixed or sparse evidence. SS=0.988, MS=0.448, RS=0.542. Safety grade: C.
- Evidence signals:
- [PASS] Literature support: SS = 0.988 (HIGH) from 59 records
- [PASS] Mechanistic plausibility: MS = 0.448 (LOW), 8 pathway(s)
- [FAIL] Safety/Risk acceptable: RS = 0.542 (MEDIUM)
- [PASS] Evidence consistency: No contradictions
- [PASS] Human clinical data: Available

Evidence Synthesis & Dimensional Assessment

1. Mechanistic Assessment

Mechanistic Assessment: LOW (Score = 0.448).

Literature Grounding: MODERATE.

Candidate Biological Mechanisms Discovered: 2.

Primary Mechanistic Chain: Drug: Aspirin → Target: PTGS2 (PTGS2 (Prostaglandin G/H synthase 2) [UniProt: P35354]) [INHIBITOR via ChEMBL] → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [PARTICIPATES_IN via Reactome] → Gene: TP53 [CONTAINS_ASSOCIATED_GENE via Reactome + Open Targets/DisGeNET] → Disease: Pancreatic Cancer [ASSOCIATED_WITH via Open Targets / DisGeNET].

Rationale: Mechanistic Score (MS) derived from candidate mechanism quality. Best candidate support level: WEAK_SPECULATIVE. Discovered 2 candidate biological mechanism(s). Literature grounding: MODERATE (7 claim(s)). Final MS = 0.448 (LOW).

2. Clinical Evidence Assessment

Clinical Evidence Assessment: HIGH Literature Support (Score = 0.988).

Clinical Trial Status: RETRIEVED.

Clinical Trial Findings: 4 trial(s) evaluated, 1 clinically significant failure(s).

Literature Support Rationale: Support Score (SS) from 59 evidence records.

Breakdown: Cell-line (in vitro): 1 records (avg ERW: 0.30); Reviews/Observational: 55 records (avg ERW: 0.62); Extracted claims: 3

Formula: $SS = 1 - \exp(-0.12 \times \text{quality_weighted_sum})$. Quality-weighted sum = 35.28 (raw ERW sum would be 37.02, capped to prevent inflation from low-quality records). Prior knowledge boost: +0.193 applied.

Final SS = 0.988 (HIGH).

3. Safety & Risk Assessment

Safety & Risk Assessment: Grade C — Moderate concerns — monitoring and caution recommended (Risk Score = 0.542, MEDIUM).

Boxed Warning: NO.

Risk Factors & Signal Analysis: Risk Score (RS) computed from clinically significant safety signals.

Trial analysis: 1 clinically significant failure(s); 0 administrative/external termination(s) excluded.

Risk factors: Safety-terminated trials: 1 [HIGH RISK]; Safety grade: C (grade penalty: 0.8).

Formula: $RS = 1 - \exp(-0.30 \times \text{raw_risk_burden})$. Raw burden = 2.60. Final RS = 0.542 (MEDIUM).

4. Final Synthesis

Synthesis & Recommendation: UNCERTAIN.

Regulatory Status: NONE (Term: aspirin → colorectal cancer).

1. Mechanistic Plausibility: LOW (0.448)

2. Literature/Clinical Support: HIGH (0.988)

3. Risk & Safety Profile: MEDIUM (0.542, Grade C)

Conclusion Summary: Rule 5 (UNCERTAIN): Mixed or sparse evidence. SS=0.988, MS=0.448, RS=0.542.

Safety grade: C.

Confidence Narrative

Overall confidence is estimated at 100% after accounting for data gaps (total penalty: -0%). Evidence

quality: The analysis is based on 59 retrieved records representing moderate-quality mixed evidence.

ERW values are capped per evidence tier to prevent inflation from low-quality records (e.g., reviews and hypotheses). Safety assessment: Grade C — Moderate concerns — monitoring and caution

recommended. No boxed warning detected. Safety grade scale: A = Excellent | B = Acceptable | C =

Caution | D = Unsafe. Prior knowledge: Related cache entry found (similarity: 0.81): Aspirin → Colorectal Cancer. Cache provides a positive evidence boost of 0.19.